$$\nabla^2_{cylind} = \frac{d^2}{d\rho^2} + \frac{1}{\rho}\frac{d}{d\rho} + \frac{1}{\rho^2}\frac{d^2}{d\varphi^2} + \frac{d^2}{dz^2} \qquad \psi(\rho\varphi z) = U(\rho)\Phi(\varphi)Z(z)$$

$$\Rightarrow -\frac{\hbar^2}{2\mu}\nabla^2\psi + \frac{eBL_z}{2\mu c}\psi + \frac{e^2B^2}{8\mu c^2}\rho^2\psi = E\psi$$

$$\Rightarrow -\frac{\hbar^2}{2\mu}\left\{\frac{d^2}{d\rho^2}+\frac{1}{\rho}\frac{d}{d\rho}+\frac{1}{\rho^2}\frac{d^2}{d\varphi^2}+\frac{d^2}{dz^2}\right\}U(\rho)\Phi(\varphi)Z(z) + \frac{eBL_z}{2\mu c}U(\rho)\Phi(\varphi)Z(z) + \frac{e^2B^2\rho^2}{8\mu c^2}U(\rho)\Phi(\varphi)Z(z) = EU(\rho)\Phi(\rho)Z(z) \qquad \frac{1}{U(\rho)\Phi(\varphi)Z(z)}$$

$$-\frac{\hbar^2}{2\mu}\left\{\frac{1}{U}\frac{d^2U}{d\rho^2}+\frac{1}{\rho U}\frac{dU}{d\rho}+\frac{1}{\rho^2}\frac{d^2\Phi}{d\varphi^2\Phi}+\frac{d^2Z}{dz^2Z}\right\} + \frac{eB}{2i\mu c}\frac{d\Phi}{d\varphi\,\Phi} + \frac{e^2B^2\rho^2}{8\mu c^2} = E$$

$$\Phi = Ce^{+im\varphi}$$

$$\frac{d\Phi}{d\varphi} = im\Phi \qquad \frac{d^2\Phi}{d\varphi^2} = -m^2\Phi$$

$$\Rightarrow -\frac{\hbar^2}{2\mu}\left\{\frac{1}{U}\frac{d^2U}{d\rho^2}+\frac{1}{\rho U}\frac{dU}{d\rho}-\frac{m^2}{\rho^2}\right\} - \frac{\hbar^2}{2\mu}\frac{d^2Z}{dz^2Z} + \frac{eBm}{2\mu c} + \frac{e^2B^2\rho^2}{8\mu c^2} = E$$

$$\Rightarrow \frac{\hbar^2}{2\mu}\left\{\frac{d^2U}{d\rho^2U}+\frac{1}{\rho U}\frac{dU}{d\rho}-\frac{m^2}{\rho^2}\right\} - \frac{\hbar^2}{2\mu}\left\{\frac{d^2Z}{dz^2Z} - \frac{eBm}{c\hbar} - \frac{e^2B^2\rho^2}{4\mu c^2\hbar^2}\right\} = E$$

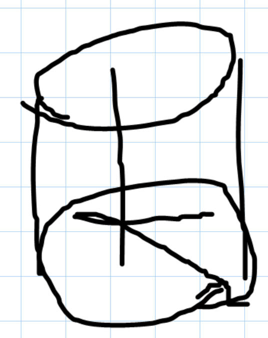
$$\Rightarrow \frac{d^2U}{Ud\rho^2} + \frac{1}{\rho U}\frac{dU}{d\rho} - \frac{m^2}{\rho^2} + \frac{d^2Z}{dz^2Z} - \frac{eBm}{c\hbar} - \frac{e^2B^2\rho^2}{4c\hbar^2} + \frac{2\mu E}{\hbar^2} = 0$$

$$\frac{d^2U}{Ud\rho^2} + \frac{1}{\rho U}\frac{dU}{d\rho} - \frac{m^2}{\rho^2} - \frac{e^2B^2\rho^2}{4c\hbar^2} - \frac{eBm}{c\hbar} + \frac{2\mu E}{\hbar^2} + \frac{d^2Z}{dz^2Z} = 0$$

$$\frac{d^2U}{d\rho^2} + \frac{1}{\rho}\cdot\frac{dU}{d\rho} - \frac{m^2U}{\rho^2} - \frac{e^2B^2\rho^2U}{4c\hbar^2} - \frac{eBmU}{c\hbar} + \frac{2\mu EU}{\hbar^2} + \frac{d^2Z\,U}{dz^2Z} = 0$$

fixed

$$x = \sqrt{\frac{eB}{2\hbar c}}\,\rho$$



$$\rho^2 = x^2 + y^2$$

$$x = \sqrt{\frac{\mu\omega_L}{\hbar}}\,\rho \qquad \sqrt{\frac{\cancel{\mu}eB}{\hbar 2\cancel{\mu}c}}\,\rho \qquad = \sqrt{\frac{eB}{2\hbar c}}\,\rho = x$$

$$\left[\frac{d^2}{d\rho^2}+\frac{1}{\rho}\frac{d}{d\rho}-\frac{m^2}{\rho^2}-\frac{e^2B^2\rho^2}{4c^2\hbar^2}\right]U + \left(\frac{2\mu E}{\hbar^2}-\frac{eBm}{\hbar c}-k_z^2\right)U = 0$$

$$\Rightarrow \left(\frac{eB}{2\hbar c}\frac{d^2}{dx^2} + \frac{eB}{2\hbar c}\frac{1}{x}\frac{d}{dx} - \frac{m^2}{x}\frac{eB}{2\hbar c} - \frac{e^2B^2}{4\hbar^2c^2}\left(\frac{2\hbar c}{eB}\right)x^2\right)U + \left(\frac{2\mu E}{\hbar^2} - \frac{eBm}{\hbar c} - k_z^2\right)U = 0$$

$$-\frac{eB}{2c\hbar}$$

$$\frac{eB}{2\hbar c}\left\{\frac{d^2}{dx^2}+\frac{1}{x}\frac{d}{dx}-\frac{m^2}{x^2}-x^2\right\}U + \frac{2\mu}{\hbar^2}\left(E - \frac{eBm\hbar}{2\mu c} - \frac{\hbar^2k_z^2}{2\mu}\right)U = 0$$

$$\frac{eB}{2\hbar c}\left\{\frac{d^2}{dx^2}+\frac{1}{x}\frac{d}{dx}-\frac{m^2}{x^2}-x^2\right\}U+\frac{2\mu}{\hbar^2}\left(E-\frac{eBm\hbar}{2\mu c}-\frac{\hbar^2 k_z^2}{2\mu}\right)U=0$$

$$\frac{eB}{2\hbar c}\left\{\frac{d^2}{dx^2}+\frac{1}{x}\frac{d}{dx}-\frac{m^2}{x^2}-x^2\right\}U+\underbrace{\left(\frac{2\mu E}{\hbar^2}-\frac{eBm}{\hbar c}-k_z^2\right)}U=0$$

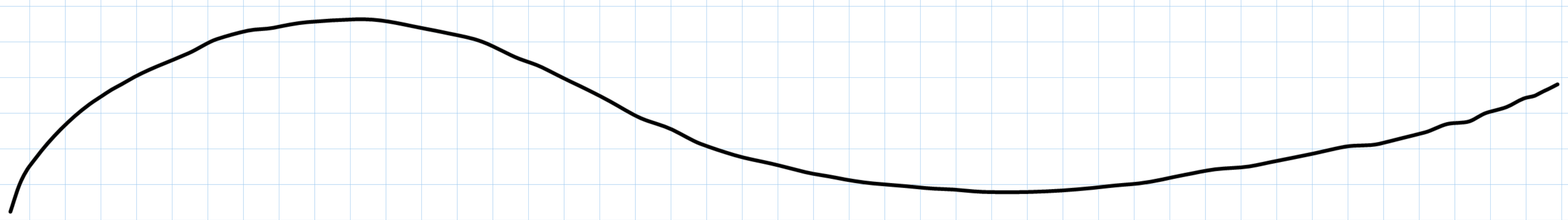
$$\frac{2\mu E}{\hbar^2}-\frac{eBm}{\hbar c}-k_z^2 \Rightarrow \frac{eB}{2\hbar c}\left\{\left(\frac{2\mu E}{\hbar^2}\right)\left(\frac{2\hbar c}{eB}\right)-\frac{eBm}{\hbar c}\left(\frac{2\hbar c}{eB}\right)-k_z^2\left(\frac{2\hbar c}{eB}\right)\right\}$$

$$\frac{eB}{2\hbar c}\left\{\frac{4\mu E c}{\hbar eB}-2m-\frac{2\hbar k_z^2 c}{eB}\right\}$$

$$\cancel{\frac{eB}{2\hbar c}}\left\{\frac{d^2}{dx^2}+\frac{1}{x}\frac{d}{dx}-\frac{m^2}{x^2}-x^2\right\}U+\cancel{\frac{eB}{2\hbar c}}\left\{\frac{4\mu E c}{\hbar eB}-2m-\frac{2\hbar k_z^2 c}{eB}\right\}U=0$$

$$\frac{4\mu c}{eB\hbar}\cdot\left(E-\frac{\hbar^2 k_z^2}{2\mu c}\right)-2m=\lambda$$

$$\left(\frac{d^2}{dx^2}+\frac{1}{x}\frac{d}{dx}-\frac{m^2}{x^2}-x^2+\lambda\right)U=0$$



$$\left(\frac{d^2}{dx^2} + \frac{1}{x}\frac{d}{dx} - \frac{m^2}{x^2} - x^2 + \lambda\right) U = 0 \qquad x \Rightarrow \infty \;\Big|\; x = 0$$

$$x \to \infty \Rightarrow \left(\frac{d^2}{dx^2} U - x^2 U\right) = 0 \quad \text{Asymt}$$

$$\Rightarrow D^2 - x^2 \Rightarrow D = \mp x \Rightarrow D = e^{\mp x^2/2}$$

$$D^2 U - x^2 U \qquad D' = \mp x e^{\mp x^2/2} \qquad D'' = \mp e^{\mp x^2/2} + x^2 e^{\mp x^2/2} \qquad \lim_{x \to \infty} e^{\frac{1}{x^2}} \cong e^{-x^2/2}$$

$$\frac{1}{\infty} = 0$$

$$x = 0 \Rightarrow \left(\frac{d^2}{dx^2} + \frac{1}{x}\frac{d}{dx} - \frac{m^2}{x^2} + \lambda\right) U \Rightarrow U(x \to 0) \cong x^{|m|} \quad \text{o rantili}$$

$$U = e^{-x^2/2} x^{|m|} G(p) =)$$

$$U' = -e^{-x^2/2} x^{|m|+1} G + |m| e^{-x^2/2} x^{|m|-1} G + e^{-x^2/2} x^{|m|} G'$$

$$U'' = +e^{-x^2/2} x^{|m|+2} G - (|m|+1) e^{-x^2/2} x^{|m|} G - e^{-x^2/2} x^{|m|+1} G' - |m| e^{-x^2/2} x^{|m|} G + (|m|-1)|m| e^{-x^2/2} x^{|m|-2} G + |m| e^{-x^2/2} G' - e^{-x^2/2} x^{|m|+1} G' + |m| e^{-x^2/2} x^{|m|-1} G' + e^{-x^2/2} x^{|m|} G''$$

$$U'' = +e^{-x^2/2} x^{|m|+2} G - (|m|+1) e^{-x^2/2} x^{|m|} G - e^{-x^2/2} x^{|m|+1} G' - |m| e^{-x^2/2} x^{|m|} G + (|m|-1)|m| e^{-x^2/2} x^{|m|-2} G + |m| e^{-x^2/2} G' - e^{-x^2/2} x^{|m|+1} G' + |m| e^{-x^2/2} x^{|m|-1} G' + e^{-x^2/2} x^{|m|} G''$$

$$U'' = e^{-x^2/2} x^{|m|} \left\{ x^2 G - (|m|+1) G - x G' - |m| G + \frac{(|m|-1)|m|}{x^2} G + |m| G' - x G' + \frac{|m|}{x} G' + G'' \right\}$$

$$U' = e^{-x^2/2} x^{|m|} \left\{ -x G + \frac{|m|}{x} G + G' \right\} \qquad U = e^{-x^2/2} x^{|m|} G$$

$$u'' = e^{-x^2/2} x^{|m|} \left\{ x^2 G - (|m|+1) G - x G' - |m| G + \frac{(|m|-1)|m|}{x^2} G + |m| G' - x G' + \frac{|m|}{x} G' + G'' \right\}$$

$$u' = e^{-x^2/2} x^{|m|} \left\{ -xG + \frac{|m|}{x} G + G' \right\} \qquad u = e^{-x^2/2} x^{|m|} G$$

$$x = 0 \Rightarrow \left(\frac{d^2}{dx^2} + \frac{1}{x}\frac{d}{dx} - \frac{m^2}{x^2} + \lambda \right) u =$$

$$x^2 G - (|m|+1) G - x G' - |m| G + \frac{(|m|-1)|m|}{x^2} G + |m| G' - x G' + \frac{|m|}{x} G' + G'' = u''$$

$$\Rightarrow G'' + G' \left(\frac{|m|}{x} + |m| - 2x \right) + G \left(\frac{(|m|-1)|m|}{x^2} - 2|m| + x^2 - 1 \right) + G' \frac{1}{x} + G \left(\frac{|m|}{x^2} - 1 \right) - \frac{m^2 G}{x^2}$$

$$G'' + G' \left(\frac{|m|}{x} + \cancel{|m|} - 2x + \frac{1}{x} \right) + G \left(\frac{|m|^2 - |m|}{x^2} - 2|m| + x^2 - 1 + \frac{|m|}{x^2} - 1 - \frac{m^2}{x^2} \right) = 0$$

$$G'' + G' \left(\frac{2|m|+1}{x} - 2x \right) + G \left(-2|m| - 2 - \lambda \right) = 0$$

$y = x^2$ dönüşümü $\quad x = \sqrt{\frac{eB}{\hbar c}}\, \rho \qquad dy = 2x\,dx \Rightarrow \frac{dy}{dx} = 2x \qquad \frac{d}{dx} = \frac{dy}{dx} \cdot \frac{d}{dy}$

$\sqrt{y} = x \Rightarrow$

$$2x \frac{d}{dy} \Rightarrow \boxed{2\sqrt{y}\frac{d}{dy} = \frac{d}{dx}}$$

$$\frac{d^2}{dx^2} = \left(2\sqrt{y}\frac{d}{dy}\right)\left(2\sqrt{y}\frac{d}{dy}\right) = 4y\frac{d^2}{dy^2} - \frac{1}{\sqrt{y}}\frac{d}{dy} - \frac{1}{\sqrt{y}}\frac{d}{dy}$$

$$4y\frac{d^2}{dy^2} - \frac{2}{\sqrt{y}}\frac{d}{dy}$$

+ Hata

$$\left(\frac{d^2}{dx^2} + \frac{1}{x}\frac{d}{dx} - \frac{m^2}{x^2} + \lambda\right)u$$

$$\frac{dy}{dx} = 2x \qquad \frac{d}{dx} = \frac{dy}{dx}\cdot\frac{d}{dy}$$

$$2x\frac{d}{dy} \Rightarrow \boxed{2\sqrt{y}\frac{d}{dy} = \frac{d}{dx}}$$

$$\frac{d^2}{dx^2} = \left(2\sqrt{y}\frac{d}{dy}\right)\left(2\sqrt{y}\frac{d}{dy}\right) = 4y\frac{d^2}{dy^2} - \frac{\cancel{\sqrt{y}}}{\cancel{\sqrt{y}}}\frac{d}{dy} - \frac{\cancel{\sqrt{y}}}{\cancel{\sqrt{y}}}\frac{d}{dy}$$

$$4y\frac{d^2}{dy^2} - 2\frac{d}{dy}$$

+ Hnd

$y = x^2$
$x = \sqrt{y}$

$$y = x^2 \quad x = \sqrt{y}$$

$$G'' + G'\left(\frac{2|m|+1}{x} - 2x\right) + G\left(-2|m| - 2 - \lambda\right) = 0$$

$$4y\frac{d^2G}{dy^2} - 2\frac{dG}{dy} + 2\sqrt{y}\frac{dG}{dy}\left(\frac{2(|m|+1)}{\sqrt{y}} - 2\sqrt{y}\right) + \left(-2|m| - 2 - \lambda\right)G = 0$$

$$4y\frac{d^2G}{dy^2} - 2\frac{dG}{dy} + 2\frac{dG}{dy}\left(2(|m|+1) - 2y\right) + \left(-2|m| - 2 - \lambda\right)G = 0$$

$$\frac{d^2G}{dy^2} - \frac{1}{2y}\frac{dG}{dy} + \frac{1}{2y}\frac{dG}{dy}\left(2(|m|+1) - 2y\right) + \frac{1}{4y}\left(-2|m| - 2 - \lambda\right)G = 0$$

$$\frac{d^2G}{dy^2} + \frac{1}{2y}\frac{dG}{dy}\left(2(|m|+\cancel{1}) - 2y - \cancel{1}\right) + \frac{1}{4y}G\left(-2(|m|+1) - \lambda\right) \qquad \lambda = -\lambda'$$

$$\Rightarrow \frac{d^2G}{dy^2} + \frac{dG}{dy}\left(\frac{m}{y} - 1\right) + \frac{G}{y}\left(-\frac{2}{4}(|m|+1) + \frac{\lambda'}{4}\right)$$

$$\underbrace{\frac{\lambda'}{4} - \frac{1}{2}(|m|+1)}_{n_r}$$

$$G = \text{Pelinom} \Rightarrow G = L_{n_r}^{|m|}(y)$$

$$G(\rho) = \sum_n a_n \rho^n$$